Tamban Fish Inspection Report

Inspection source: WIN_20260823_20_49_53_Pro.jpg

Inspection dates: 2026-11-01T00:20:20.811Z

Export filter: All · Unique fish: 7 · Generated: 11/1/2026, 7:37:47 AM

Class distribution

Final classification	Unique fish
Class A	0
Class B	2
Class C	0
Rejected	0
Ungraded	5

Grading methodology

Model 1 locates each fish. Model 2 provides Body / Head / Tail classification evidence with original weights 50% / 30% / 20%. The recorded policy determines effective weights and acceptance safeguards. Validated Class C geometric support is bounded and applied once after weighted classification. Silhouette measurements report rotation-aware pixel length and associated Body width when geometry passes reliability checks. Bounding-box width remains a separate diagnostic proxy. Silhouette measurements remain excluded from grading; uncalibrated or unreliable width cannot contribute. Detection confidence, part confidence, weighted support and 0–10 geometric points are separate quantities.

Class B (2)

Fish #1 · Class B · Final support 54.2%

A. Grading Evidence

Evidence	Value
Model 1 detection	78.4%
Model 2 Body	Class B · 69.8% part confidence
Model 2 Head	Class B · 64.4% part confidence
Model 2 Tail	Class A · 33.0% part confidence
Effective weights	Head: 0.3 · Body: 0.5 · Tail: 0.2
Original weighted classification	Class B
Original weighted support	54.2%
Final combined support	54.2%
Ungraded reason	—

Class	Original weighted support	Final combined support
Class A	0.0660010278224945	0.0660010278224945
Class B	0.5421848714351654	0.5421848714351654
Class C	0	0
Rejected	0	0

Model 2 label	Region	Class	Confidence (0–1)	Original bbox (px)
Grade_B_Body	Body	Class B	0.6981277465820312	[519.56,1115.85,1101.5,1792.08]
Grade_B_Head	Head	Class B	0.6437366604804993	[551.47,751.48,868.75,1077.29]
Grade_A_Tail	Tail	Class A	0.33000513911247253	[447,1752.04,892.39,2160]

B. Body Width and Geometric Analysis

Measurement	Value
Measured fish length (px)	Unavailable
Maximum body width (px)	Unavailable
Relative body width ratio	Unavailable
Fish measurement status	unreliable
Fish measurement reliability	Unreliable: parent association uncertain; parent touches frame boundary; per fish segmentation unavailable; historical image unavailable
Measurement validation	Not validated for anatomical measurement or width grading
Body measurement source	Unavailable
Fish measurement method	segmentation centerline profile v1
Fish orientation (degrees)	Unavailable
Physical calibration	Not calibrated: physical calibration not validated
Silhouette measurement used for grading	No
Whole-fish bounding box	672.3196 x 1511.4575 px
Body bounding box	581.94 x 676.23 px
Whole-fish measured ROI	673 x 1512 px
Body measured ROI	583 x 678 px
Relative width proxy	0.3856
Proxy measurement source	existing ROI measurements
Measurement status	Unreliable
Orientation	vertical
Measurement reliability	Unreliable: overlapping fish; clipped fish
Calibration status	Not Calibrated
Normal reference range	Not calibrated
Width abnormality	Unreliable
Class C supporting points	Not calibrated
Contributed to final grade	No: not calibrated
Actual combined-score contribution	0

C. Image Features and Measurements

Whole Fish ROI — Measured

Recorded field	Value
bbox	[447.4203796386719,648.5425415039062,1119.739990234375,2160]

Recorded field	Value
status	measured
measurement_source	Model 1 fish crop rectangular ROI; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,0,673,1512]}]
values.mean_hue_deg	245.0723526039389
values.median_hue_deg	231.42857142857144
values.mean_saturation	0.5956559525578725
values.median_saturation	0.6808510638297872
values.mean_value	0.1774871524747701
values.median_value	0.0784313725490196
values.yellow_ratio_proxy	0.08686230807330361
values.brown_ratio_proxy	0.10210834375024568
values.dark_patch_ratio_proxy	0.6657900736652594
values.discoloration_coverage_proxy	0.7713124130286092
values.specular_ratio_proxy	0

Body ROI — Measured

Recorded field	Value
bbox	[519.56,1115.85,1101.5,1792.08]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[72,467,655,1145]}]
values.mean_hue_deg	246.09864434300965
values.median_hue_deg	225
values.mean_saturation	0.4941902998924152
values.median_saturation	0.3617021276595745
values.mean_value	0.22544743596573913
values.median_value	0.21176470588235294
values.yellow_ratio_proxy	0.08397466061516821
values.brown_ratio_proxy	0.09023108021271321
values.dark_patch_ratio_proxy	0.5421201495671357
values.discoloration_coverage_proxy	0.6432930068762428
values.specular_ratio_proxy	0

Head ROI — Measured

Recorded field	Value
bbox	[551.47,751.48,868.75,1077.29]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[104,103,422,430]}]
values.mean_hue_deg	295.682973626724
values.median_hue_deg	48.00000000000001
values.mean_saturation	0.607792590811306
values.median_saturation	0.5833333333333334
values.mean_value	0.18395119554178296
values.median_value	0.13333333333333333
values.yellow_ratio_proxy	0.20624891812359356
values.brown_ratio_proxy	0.2370511414998173
values.dark_patch_ratio_proxy	0.6684457523128113
values.discoloration_coverage_proxy	0.9257111534244995
values.specular_ratio_proxy	0

Tail ROI — Measured

Recorded field	Value
bbox	[447,1752.04,892.39,2160]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,1104,446,1512]}]
values.mean_hue_deg	240.63953205376367
values.median_hue_deg	240
values.mean_saturation	0.617330482176029
values.median_saturation	1
values.mean_value	0.16357827620628837
values.median_value	0.047058823529411764
values.yellow_ratio_proxy	0.04367800932032006
values.brown_ratio_proxy	0.01684911632814561
values.dark_patch_ratio_proxy	0.6586872417128287
values.discoloration_coverage_proxy	0.7020739910313901
values.specular_ratio_proxy	0

D. Final Grading Explanation

Decision stage	Result
Original weighted grade	Class B
Geometric support	Not calibrated
Geometric contribution	not calibrated
Final combined AI grade	Class B
Width changed classification	No
Decision reason	The top weighted grade passed all evidence gates with 54.22% support.

Model, source and threshold configuration

Recorded field	Value
session_id	Not recorded
timestamp	2026-11-01T00:20:20.811Z
traceability.model2_id	yolo26m_partdet
traceability.model2_name	YOLO26m Part Detector (r03 devbest)
traceability.model2_checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt
traceability.model2_checkpoint_sha256	f74cd336b6ba
traceability.grading_config_version	4.0
traceability.part_weights.Body	0.5
traceability.part_weights.Head	0.3
traceability.part_weights.Tail	0.2
traceability.model2_detector_threshold	0.25
traceability.minimum_part_evidence_threshold	0.25
traceability.minimum_grade_margin	0
traceability.quality_threshold	0.25
traceability.final_verdict_threshold	0.5
traceability.minimum_original_weight_coverage	0.5
traceability.grading_mode	tail_optional_v1
traceability.quality_inference_mode	full_frame
traceability.roi_padding	0
traceability.model2_available	true
traceability.model1_checkpoint	model1_fish_parent_detector.pt
traceability.model1_checkpoint_sha256	87498c748cec
traceability.detection_threshold	0.5
grading_config.config_version	4.0
grading_config.part_weights.Head	0.3
grading_config.part_weights.Body	0.5
grading_config.part_weights.Tail	0.2
grading_config.require_body	true
grading_config.minimum_part_confidence	0.25

Recorded field	Value
grading_config.minimum_final_score	Not recorded
grading_config.deprecated_minimum_final_score_used	false
grading_config.final_verdict_threshold	0.5
grading_config.active_final_verdict_threshold	0.5
grading_config.grading_mode	tail_optional_v1
grading_config.minimum_regions_observed	2
grading_config.minimum_original_weight_coverage	0.5
grading_config.strict_minimum_regions_observed	3
grading_config.strict_minimum_original_weight_coverage	1
grading_config.minimum_grade_margin	0
grading_config.minimum_track_observations	2
grading_config.temporal_evidence_limit	120
grading_config.maximum_grade_stddev	Not recorded
grading_config.rejected_override_threshold	Not recorded
grading_config.color_adjustments_enabled	false
grading_config.defect_adjustments_enabled	false
grading_config.image_diagnostics_enabled	true
grading_config.frame_quality.use_frame_quality_filter	false
grading_config.frame_quality.minimum_detection_confidence	Not recorded
grading_config.frame_quality.minimum_crop_width	Not recorded
grading_config.frame_quality.minimum_crop_height	Not recorded
grading_config.frame_quality.minimum_crop_area	Not recorded
grading_config.frame_quality.reject_clipped_crops	false
grading_config.frame_quality.sharpness_filter_enabled	false
grading_config.frame_quality.minimum_sharpness	Not recorded
grading_config.best_frame.detection_weight	0.35
grading_config.best_frame.area_weight	0.2
grading_config.best_frame.sharpness_weight	0.25
grading_config.best_frame.region_weight	0.2
grading_config.best_frame.area_reference	Not recorded
grading_config.best_frame.sharpness_reference	Not recorded
grading_config.best_frame.only_usable	true
grading_config.best_frame.clipped_penalty	0
grading_config.best_frame.save_best_fish_crop	false
grading_config.best_frame.save_annotated_best_fish_crop	false
grading_config.best_frame.best_crop_directory	results/best_fish_crops
width_threshold_configuration.scoring_enabled	false
width_threshold_configuration.calibration_validated	false

Recorded field	Value
width_threshold_configuration.calibration_id	Not recorded
width_threshold_configuration.validation_sample_count	0
width_threshold_configuration.normal_reference_min	Not recorded
width_threshold_configuration.normal_reference_max	Not recorded
width_threshold_configuration.extreme_reference	Not recorded
width_threshold_configuration.scoring_method	post_weighted_v2
width_threshold_configuration.maximum_geometric_contribution	0.1
width_threshold_configuration.maximum_body_score_shift	0.1
width_threshold_configuration.minimum_class_c_evidence	0.25
width_threshold_configuration.geometry_max_axis_deviation_degrees	15
width_threshold_configuration.geometry_min_elongation	4
width_threshold_configuration.minimum_detection_confidence	0.25
width_threshold_configuration.minimum_part_inside_fraction	0.95

Fish #3 · Class B · Final support 57.1%

A. Grading Evidence

Evidence	Value
Model 1 detection	72.6%
Model 2 Body	Class B · 62.4% part confidence
Model 2 Head	Class B · 68.9% part confidence
Model 2 Tail	Class B · 25.9% part confidence
Effective weights	Head: 0.3 · Body: 0.5 · Tail: 0.2
Original weighted classification	Class B
Original weighted support	57.1%
Final combined support	57.1%
Ungraded reason	—

Class	Original weighted support	Final combined support
Class A	0	0
Class B	0.5705721914768219	0.5705721914768219
Class C	0	0
Rejected	0	0

Model 2 label	Region	Class	Confidence (0–1)	Original bbox (px)
Grade_B_Head	Head	Class B	0.6889662742614746	[1109.86,282,1459.81,640.17]
Grade_B_Body	Body	Class B	0.6240850687026978	[1006.05,736.82,1499,1616.11]
Grade_B_Tail	Tail	Class B	0.2591988742351532	[997,1483.73,1374.07,2124.94]

B. Body Width and Geometric Analysis

Measurement	Value
Measured fish length (px)	Unavailable
Maximum body width (px)	Unavailable
Relative body width ratio	Unavailable
Fish measurement status	unavailable
Fish measurement reliability	Unavailable: parent association uncertain; per fish segmentation unavailable; historical image unavailable
Measurement validation	Not validated for anatomical measurement or width grading
Body measurement source	Unavailable
Fish measurement method	segmentation centerline profile v1
Fish orientation (degrees)	Unavailable
Physical calibration	Not calibrated: physical calibration not validated
Silhouette measurement used for grading	No

Measurement	Value
Whole-fish bounding box	501.4095 x 1861.1524 px
Body bounding box	492.95 x 879.29 px
Whole-fish measured ROI	502 x 1862 px
Body measured ROI	493 x 881 px
Relative width proxy	0.2648
Proxy measurement source	existing ROI measurements
Measurement status	Unreliable
Orientation	vertical
Measurement reliability	Unreliable: overlapping fish; clipped fish
Calibration status	Not Calibrated
Normal reference range	Not calibrated
Width abnormality	Unreliable
Class C supporting points	Not calibrated
Contributed to final grade	No: not calibrated
Actual combined-score contribution	0

C. Image Features and Measurements

Whole Fish ROI — Measured

Recorded field	Value
bbox	[997.0737915039062,282.08099365234375,1498.4832763671875,2143.2333984375]
status	measured
measurement_source	Model 1 fish crop rectangular ROI; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,0,502,1862]}]
values.mean_hue_deg	249.0074507249867
values.median_hue_deg	110.00000000000003
values.mean_saturation	0.4042466011750104
values.median_saturation	0.2564102564102565
values.mean_value	0.25634973217636814
values.median_value	0.29411764705882354
values.yellow_ratio_proxy	0.15854519622904728
values.brown_ratio_proxy	0.11179663729614303
values.dark_patch_ratio_proxy	0.45850004921238785
values.discoloration_coverage_proxy	0.6270449886811508
values.specular_ratio_proxy	0

Body ROI — Measured

Recorded field	Value
bbox	[1006.05,736.82,1499,1616.11]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[9,454,502,1335]}]
values.mean_hue_deg	351.6071149498801
values.median_hue_deg	90
values.mean_saturation	0.357169400913011
values.median_saturation	0.23076923076923084
values.mean_value	0.2948469149202092
values.median_value	0.34509803921568627
values.yellow_ratio_proxy	0.1875450403262013
values.brown_ratio_proxy	0.15021423654200808
values.dark_patch_ratio_proxy	0.3656595285184409
values.discoloration_coverage_proxy	0.5684716565400281
values.specular_ratio_proxy	0

Head ROI — Measured

Recorded field	Value
bbox	[1109.86,282,1459.81,640.17]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[112,0,463,359]}]
values.mean_hue_deg	248.24871359025488
values.median_hue_deg	60
values.mean_saturation	0.49731374593625466
values.median_saturation	0.4117647058823529
values.mean_value	0.20119695776476595
values.median_value	0.17647058823529413
values.yellow_ratio_proxy	0.27731352522438873
values.brown_ratio_proxy	0.12374512931615995
values.dark_patch_ratio_proxy	0.5994254378655492
values.discoloration_coverage_proxy	0.8804767913403011
values.specular_ratio_proxy	0

Tail ROI — Measured

Recorded field	Value
bbox	[997,1483.73,1374.07,2124.94]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,1201,378,1843]}]
values.mean_hue_deg	236.52600039645307
values.median_hue_deg	165.00000000000006
values.mean_saturation	0.3782852916402867
values.median_saturation	0.16091954022988514
values.mean_value	0.24145603320363568
values.median_value	0.2980392156862745
values.yellow_ratio_proxy	0.07826072623580412
values.brown_ratio_proxy	0.034560483937430976
values.dark_patch_ratio_proxy	0.45166806771168144
values.discoloration_coverage_proxy	0.5310578714005505
values.specular_ratio_proxy	0

D. Final Grading Explanation

Decision stage	Result
Original weighted grade	Class B
Geometric support	Not calibrated
Geometric contribution	not calibrated
Final combined AI grade	Class B
Width changed classification	No
Decision reason	The top weighted grade passed all evidence gates with 57.06% support.

Model, source and threshold configuration

Recorded field	Value
session_id	Not recorded
timestamp	2026-11-01T00:20:20.811Z
traceability.model2_id	yolo26m_partdet
traceability.model2_name	YOLO26m Part Detector (r03 devbest)
traceability.model2_checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt
traceability.model2_checkpoint_sha256	f74cd336b6ba
traceability.grading_config_version	4.0
traceability.part_weights.Body	0.5
traceability.part_weights.Head	0.3

Recorded field	Value
traceability.part_weights.Tail	0.2
traceability.model2_detector_threshold	0.25
traceability.minimum_part_evidence_threshold	0.25
traceability.minimum_grade_margin	0
traceability.quality_threshold	0.25
traceability.final_verdict_threshold	0.5
traceability.minimum_original_weight_coverage	0.5
traceability.grading_mode	tail_optional_v1
traceability.quality_inference_mode	full_frame
traceability.roi_padding	0
traceability.model2_available	true
traceability.model1_checkpoint	model1_fish_parent_detector.pt
traceability.model1_checkpoint_sha256	87498c748cec
traceability.detection_threshold	0.5
grading_config.config_version	4.0
grading_config.part_weights.Head	0.3
grading_config.part_weights.Body	0.5
grading_config.part_weights.Tail	0.2
grading_config.require_body	true
grading_config.minimum_part_confidence	0.25
grading_config.minimum_final_score	Not recorded
grading_config.deprecated_minimum_final_score_used	false
grading_config.final_verdict_threshold	0.5
grading_config.active_final_verdict_threshold	0.5
grading_config.grading_mode	tail_optional_v1
grading_config.minimum_regions_observed	2
grading_config.minimum_original_weight_coverage	0.5
grading_config.strict_minimum_regions_observed	3
grading_config.strict_minimum_original_weight_coverage	1
grading_config.minimum_grade_margin	0
grading_config.minimum_track_observations	2
grading_config.temporal_evidence_limit	120
grading_config.maximum_grade_stddev	Not recorded
grading_config.rejected_override_threshold	Not recorded
grading_config.color_adjustments_enabled	false
grading_config.defect_adjustments_enabled	false
grading_config.image_diagnostics_enabled	true
grading_config.frame_quality.use_frame_quality_filter	false

Recorded field	Value
grading_config.frame_quality.minimum_detection_confidence	Not recorded
grading_config.frame_quality.minimum_crop_width	Not recorded
grading_config.frame_quality.minimum_crop_height	Not recorded
grading_config.frame_quality.minimum_crop_area	Not recorded
grading_config.frame_quality.reject_clipped_crops	false
grading_config.frame_quality.sharpness_filter_enabled	false
grading_config.frame_quality.minimum_sharpness	Not recorded
grading_config.best_frame.detection_weight	0.35
grading_config.best_frame.area_weight	0.2
grading_config.best_frame.sharpness_weight	0.25
grading_config.best_frame.region_weight	0.2
grading_config.best_frame.area_reference	Not recorded
grading_config.best_frame.sharpness_reference	Not recorded
grading_config.best_frame.only_usable	true
grading_config.best_frame.clipped_penalty	0
grading_config.best_frame.save_best_fish_crop	false
grading_config.best_frame.save_annotated_best_fish_crop	false
grading_config.best_frame.best_crop_directory	results/best_fish_crops
width_threshold_configuration.scoring_enabled	false
width_threshold_configuration.calibration_validated	false
width_threshold_configuration.calibration_id	Not recorded
width_threshold_configuration.validation_sample_count	0
width_threshold_configuration.normal_reference_min	Not recorded
width_threshold_configuration.normal_reference_max	Not recorded
width_threshold_configuration.extreme_reference	Not recorded
width_threshold_configuration.scoring_method	post_weighted_v2
width_threshold_configuration.maximum_geometric_contribution	0.1
width_threshold_configuration.maximum_body_score_shift	0.1
width_threshold_configuration.minimum_class_c_evidence	0.25
width_threshold_configuration.geometry_max_axis_deviation_degrees	15
width_threshold_configuration.geometry_min_elongation	4
width_threshold_configuration.minimum_detection_confidence	0.25
width_threshold_configuration.minimum_part_inside_fraction	0.95

Ungraded (5)

Fish #2 · Ungraded · Final support 21.1%

A. Grading Evidence

Evidence	Value
Model 1 detection	74.1%
Model 2 Body	Class B · 42.1% part confidence
Model 2 Head	Not Detected
Model 2 Tail	Not Detected
Effective weights	Head: 0 · Body: 0.5 · Tail: 0
Original weighted classification	Ungraded
Original weighted support	21.1%
Final combined support	21.1%
Ungraded reason	Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Highest weighted grade support was 21.05%, below the 50.00% final verdict threshold.

Class	Original weighted support	Final combined support
Class A	0	0
Class B	0.21053625643253326	0.21053625643253326
Class C	0	0
Rejected	0	0

Model 2 label	Region	Class	Confidence (0–1)	Original bbox (px)
Grade_B_Body	Body	Class B	0.42107251286506653	[2839.49,22.32,3668.61,635.96]

B. Body Width and Geometric Analysis

Measurement	Value
Measured fish length (px)	Unavailable
Maximum body width (px)	Unavailable
Relative body width ratio	Unavailable
Fish measurement status	unreliable
Fish measurement reliability	Unreliable: parent association uncertain; parent touches frame boundary; per fish segmentation unavailable; historical image unavailable
Measurement validation	Not validated for anatomical measurement or width grading
Body measurement source	Unavailable
Fish measurement method	segmentation centerline profile v1
Fish orientation (degrees)	Unavailable
Physical calibration	Not calibrated: physical calibration not validated
Silhouette measurement used for grading	No

Measurement	Value
Whole-fish bounding box	1276.158 x 748.4773 px
Body bounding box	829.12 x 613.64 px
Whole-fish measured ROI	1277 x 749 px
Body measured ROI	830 x 614 px
Relative width proxy	0.4808
Proxy measurement source	existing ROI measurements
Measurement status	Unreliable
Orientation	unknown
Measurement reliability	Unreliable: overlapping fish; clipped fish; orientation unconfirmed
Calibration status	Not Calibrated
Normal reference range	Not calibrated
Width abnormality	Unreliable
Class C supporting points	Not calibrated
Contributed to final grade	No: not calibrated
Actual combined-score contribution	0

C. Image Features and Measurements

Whole Fish ROI — Measured

Recorded field	Value
bbox	[2563.842041015625,0,3840,748.477294921875]
status	measured
measurement_source	Model 1 fish crop rectangular ROI; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,0,1277,749]}]
values.mean_hue_deg	243.01669167764345
values.median_hue_deg	222
values.mean_saturation	0.48741388408533165
values.median_saturation	0.3116883116883116
values.mean_value	0.21441533880511127
values.median_value	0.2
values.yellow_ratio_proxy	0.08203472549669463
values.brown_ratio_proxy	0.0777857817209686
values.dark_patch_ratio_proxy	0.5741709384373631
values.discoloration_coverage_proxy	0.6598356670810362
values.specular_ratio_proxy	0

Body ROI — Measured

Recorded field	Value
bbox	[2839.49,22.32,3668.61,635.96]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[276,22,1106,636]}]
values.mean_hue_deg	239.30428955928784
values.median_hue_deg	224
values.mean_saturation	0.4615062860189333
values.median_saturation	0.26562499999999994
values.mean_value	0.21976049051542443
values.median_value	0.2235294117647059
values.yellow_ratio_proxy	0.08053059142105883
values.brown_ratio_proxy	0.05785291001138103
values.dark_patch_ratio_proxy	0.5451355912248342
values.discoloration_coverage_proxy	0.6306561751893568
values.specular_ratio_proxy	0

Head ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded

Recorded field	Value
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Tail ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded

Recorded field	Value
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

D. Final Grading Explanation

Decision stage	Result
Original weighted grade	Ungraded
Geometric support	Not calibrated
Geometric contribution	not calibrated
Final combined AI grade	Ungraded
Width changed classification	No
Decision reason	Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Highest weighted grade support was 21.05%, below the 50.00% final verdict threshold.
Next step	Review the reason above and acquire a clear view with reliable Model 2 Body evidence; geometric support cannot bypass missing evidence.

Model, source and threshold configuration

Recorded field	Value
session_id	Not recorded
timestamp	2026-11-01T00:20:20.811Z
traceability.model2_id	yolo26m_partdet
traceability.model2_name	YOLO26m Part Detector (r03 devbest)
traceability.model2_checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt
traceability.model2_checkpoint_sha256	f74cd336b6ba
traceability.grading_config_version	4.0
traceability.part_weights.Body	0.5
traceability.part_weights.Head	0.3
traceability.part_weights.Tail	0.2
traceability.model2_detector_threshold	0.25
traceability.minimum_part_evidence_threshold	0.25
traceability.minimum_grade_margin	0
traceability.quality_threshold	0.25
traceability.final_verdict_threshold	0.5
traceability.minimum_original_weight_coverage	0.5
traceability.grading_mode	tail_optional_v1
traceability.quality_inference_mode	full_frame
traceability.roi_padding	0
traceability.model2_available	true
traceability.model1_checkpoint	model1_fish_parent_detector.pt
traceability.model1_checkpoint_sha256	87498c748cec

Recorded field	Value
traceability.detection_threshold	0.5
grading_config.config_version	4.0
grading_config.part_weights.Head	0.3
grading_config.part_weights.Body	0.5
grading_config.part_weights.Tail	0.2
grading_config.require_body	true
grading_config.minimum_part_confidence	0.25
grading_config.minimum_final_score	Not recorded
grading_config.deprecated_minimum_final_score_used	false
grading_config.final_verdict_threshold	0.5
grading_config.active_final_verdict_threshold	0.5
grading_config.grading_mode	tail_optional_v1
grading_config.minimum_regions_observed	2
grading_config.minimum_original_weight_coverage	0.5
grading_config.strict_minimum_regions_observed	3
grading_config.strict_minimum_original_weight_coverage	1
grading_config.minimum_grade_margin	0
grading_config.minimum_track_observations	2
grading_config.temporal_evidence_limit	120
grading_config.maximum_grade_stddev	Not recorded
grading_config.rejected_override_threshold	Not recorded
grading_config.color_adjustments_enabled	false
grading_config.defect_adjustments_enabled	false
grading_config.image_diagnostics_enabled	true
grading_config.frame_quality.use_frame_quality_filter	false
grading_config.frame_quality.minimum_detection_confidence	Not recorded
grading_config.frame_quality.minimum_crop_width	Not recorded
grading_config.frame_quality.minimum_crop_height	Not recorded
grading_config.frame_quality.minimum_crop_area	Not recorded
grading_config.frame_quality.reject_clipped_crops	false
grading_config.frame_quality.sharpness_filter_enabled	false
grading_config.frame_quality.minimum_sharpness	Not recorded
grading_config.best_frame.detection_weight	0.35
grading_config.best_frame.area_weight	0.2
grading_config.best_frame.sharpness_weight	0.25
grading_config.best_frame.region_weight	0.2
grading_config.best_frame.area_reference	Not recorded
grading_config.best_frame.sharpness_reference	Not recorded

Recorded field	Value
grading_config.best_frame.only_usable	true
grading_config.best_frame.clipped_penalty	0
grading_config.best_frame.save_best_fish_crop	false
grading_config.best_frame.save_annotated_best_fish_crop	false
grading_config.best_frame.best_crop_directory	results/best_fish_crops
width_threshold_configuration.scoring_enabled	false
width_threshold_configuration.calibration_validated	false
width_threshold_configuration.calibration_id	Not recorded
width_threshold_configuration.validation_sample_count	0
width_threshold_configuration.normal_reference_min	Not recorded
width_threshold_configuration.normal_reference_max	Not recorded
width_threshold_configuration.extreme_reference	Not recorded
width_threshold_configuration.scoring_method	post_weighted_v2
width_threshold_configuration.maximum_geometric_contribution	0.1
width_threshold_configuration.maximum_body_score_shift	0.1
width_threshold_configuration.minimum_class_c_evidence	0.25
width_threshold_configuration.geometry_max_axis_deviation_degrees	15
width_threshold_configuration.geometry_min_elongation	4
width_threshold_configuration.minimum_detection_confidence	0.25
width_threshold_configuration.minimum_part_inside_fraction	0.95

Fish #4 · Ungraded · Final support -

A. Grading Evidence

Evidence	Value
Model 1 detection	64.9%
Model 2 Body	Not Detected
Model 2 Head	Not Detected
Model 2 Tail	Not Detected
Effective weights	Head: 0 · Body: 0 · Tail: 0
Original weighted classification	Ungraded
Original weighted support	-
Final combined support	-
Ungraded reason	Model 2 part-to-parent association is ambiguous, so no verdict was issued. Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Reliable original anatomy-weight coverage was below 50.00%. Highest weighted grade support was 0.00%, below the 50.00% final verdict threshold.

Class	Original weighted support	Final combined support
Class A	0	0
Class B	0	0
Class C	0	0
Rejected	0	0

Model 2 detections: Not Detected

B. Body Width and Geometric Analysis

Measurement	Value
Measured fish length (px)	Unavailable
Maximum body width (px)	Unavailable
Relative body width ratio	Unavailable
Fish measurement status	unavailable
Fish measurement reliability	Unavailable: parent association uncertain; per fish segmentation unavailable; historical image unavailable
Measurement validation	Not validated for anatomical measurement or width grading
Body measurement source	Unavailable
Fish measurement method	segmentation centerline profile v1
Fish orientation (degrees)	Unavailable
Physical calibration	Not calibrated: physical calibration not validated
Silhouette measurement used for grading	No
Whole-fish bounding box	622.4828 x 1041.8015 px
Body bounding box	Unavailable

Measurement	Value
Whole-fish measured ROI	623 x 1043 px
Body measured ROI	Unavailable
Relative width proxy	Unavailable
Proxy measurement source	detector bounding boxes
Measurement status	Missing Detection
Orientation	unknown
Measurement reliability	Unavailable: body evidence unavailable; association unreliable; overlapping fish
Calibration status	Not Calibrated
Normal reference range	Not calibrated
Width abnormality	Unavailable
Class C supporting points	Not calibrated
Contributed to final grade	No: not calibrated
Actual combined-score contribution	0

C. Image Features and Measurements

Whole Fish ROI — Measured

Recorded field	Value
bbox	[1190.229248046875,1114.48974609375,1812.7120361328125,2156.291259765625]
status	measured
measurement_source	Model 1 fish crop rectangular ROI; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,0,623,1043]}]
values.mean_hue_deg	224.33104625190936
values.median_hue_deg	120
values.mean_saturation	0.43875696371466255
values.median_saturation	0.24999999999999992
values.mean_value	0.24810398331717878
values.median_value	0.3058823529411765
values.yellow_ratio_proxy	0.14738014955624057
values.brown_ratio_proxy	0.06648773678840363
values.dark_patch_ratio_proxy	0.4511956958335706
values.discoloration_coverage_proxy	0.5984481116177713
values.specular_ratio_proxy	0

Body ROI — Not Detected

Recorded field	Value
bbox	Not recorded

Recorded field	Value
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Head ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured

Recorded field	Value
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Tail ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded

Recorded field	Value
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

D. Final Grading Explanation

Decision stage	Result
Original weighted grade	Ungraded
Geometric support	Not calibrated
Geometric contribution	not calibrated
Final combined AI grade	Ungraded
Width changed classification	No
Decision reason	Model 2 part-to-parent association is ambiguous, so no verdict was issued. Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Reliable original anatomy-weight coverage was below 50.00%. Highest weighted grade support was 0.00%, below the 50.00% final verdict threshold.
Next step	Review the reason above and acquire a clear view with reliable Model 2 Body evidence; geometric support cannot bypass missing evidence.

Model, source and threshold configuration

Recorded field	Value
session_id	Not recorded
timestamp	2026-11-01T00:20:20.811Z
traceability.model2_id	yolo26m_partdet
traceability.model2_name	YOLO26m Part Detector (r03 devbest)
traceability.model2_checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt
traceability.model2_checkpoint_sha256	f74cd336b6ba
traceability.grading_config_version	4.0
traceability.part_weights.Body	0.5
traceability.part_weights.Head	0.3
traceability.part_weights.Tail	0.2
traceability.model2_detector_threshold	0.25
traceability.minimum_part_evidence_threshold	0.25
traceability.minimum_grade_margin	0

Recorded field	Value
traceability.quality_threshold	0.25
traceability.final_verdict_threshold	0.5
traceability.minimum_original_weight_coverage	0.5
traceability.grading_mode	tail_optional_v1
traceability.quality_inference_mode	full_frame
traceability.roi_padding	0
traceability.model2_available	true
traceability.model1_checkpoint	model1_fish_parent_detector.pt
traceability.model1_checkpoint_sha256	87498c748cec
traceability.detection_threshold	0.5
grading_config.config_version	4.0
grading_config.part_weights.Head	0.3
grading_config.part_weights.Body	0.5
grading_config.part_weights.Tail	0.2
grading_config.require_body	true
grading_config.minimum_part_confidence	0.25
grading_config.minimum_final_score	Not recorded
grading_config.deprecated_minimum_final_score_used	false
grading_config.final_verdict_threshold	0.5
grading_config.active_final_verdict_threshold	0.5
grading_config.grading_mode	tail_optional_v1
grading_config.minimum_regions_observed	2
grading_config.minimum_original_weight_coverage	0.5
grading_config.strict_minimum_regions_observed	3
grading_config.strict_minimum_original_weight_coverage	1
grading_config.minimum_grade_margin	0
grading_config.minimum_track_observations	2
grading_config.temporal_evidence_limit	120
grading_config.maximum_grade_stddev	Not recorded
grading_config.rejected_override_threshold	Not recorded
grading_config.color_adjustments_enabled	false
grading_config.defect_adjustments_enabled	false
grading_config.image_diagnostics_enabled	true
grading_config.frame_quality.use_frame_quality_filter	false
grading_config.frame_quality.minimum_detection_confidence	Not recorded
grading_config.frame_quality.minimum_crop_width	Not recorded
grading_config.frame_quality.minimum_crop_height	Not recorded
grading_config.frame_quality.minimum_crop_area	Not recorded

Recorded field	Value
grading_config.frame_quality.reject_clipped_crops	false
grading_config.frame_quality.sharpness_filter_enabled	false
grading_config.frame_quality.minimum_sharpness	Not recorded
grading_config.best_frame.detection_weight	0.35
grading_config.best_frame.area_weight	0.2
grading_config.best_frame.sharpness_weight	0.25
grading_config.best_frame.region_weight	0.2
grading_config.best_frame.area_reference	Not recorded
grading_config.best_frame.sharpness_reference	Not recorded
grading_config.best_frame.only_usable	true
grading_config.best_frame.clipped_penalty	0
grading_config.best_frame.save_best_fish_crop	false
grading_config.best_frame.save_annotated_best_fish_crop	false
grading_config.best_frame.best_crop_directory	results/best_fish_crops
width_threshold_configuration.scoring_enabled	false
width_threshold_configuration.calibration_validated	false
width_threshold_configuration.calibration_id	Not recorded
width_threshold_configuration.validation_sample_count	0
width_threshold_configuration.normal_reference_min	Not recorded
width_threshold_configuration.normal_reference_max	Not recorded
width_threshold_configuration.extreme_reference	Not recorded
width_threshold_configuration.scoring_method	post_weighted_v2
width_threshold_configuration.maximum_geometric_contribution	0.1
width_threshold_configuration.maximum_body_score_shift	0.1
width_threshold_configuration.minimum_class_c_evidence	0.25
width_threshold_configuration.geometry_max_axis_deviation_degrees	15
width_threshold_configuration.geometry_min_elongation	4
width_threshold_configuration.minimum_detection_confidence	0.25
width_threshold_configuration.minimum_part_inside_fraction	0.95

Fish #5 · Ungraded · Final support 38.0%

A. Grading Evidence

Evidence	Value
Model 1 detection	64.5%
Model 2 Body	Class B · 75.9% part confidence
Model 2 Head	Not Detected
Model 2 Tail	Not Detected
Effective weights	Head: 0 · Body: 0.5 · Tail: 0
Original weighted classification	Ungraded
Original weighted support	38.0%
Final combined support	38.0%
Ungraded reason	Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Highest weighted grade support was 37.95%, below the 50.00% final verdict threshold.

Class	Original weighted support	Final combined support
Class A	0	0
Class B	0.3795393109321594	0.3795393109321594
Class C	0	0
Rejected	0	0

Model 2 label	Region	Class	Confidence (0–1)	Original bbox (px)
Grade_B_Body	Body	Class B	0.7590786218643188	[2193.06,489,3161.15,973.25]

B. Body Width and Geometric Analysis

Measurement	Value
Measured fish length (px)	Unavailable
Maximum body width (px)	Unavailable
Relative body width ratio	Unavailable
Fish measurement status	unavailable
Fish measurement reliability	Unavailable: parent association uncertain; per fish segmentation unavailable; historical image unavailable
Measurement validation	Not validated for anatomical measurement or width grading
Body measurement source	Unavailable
Fish measurement method	segmentation centerline profile v1
Fish orientation (degrees)	Unavailable
Physical calibration	Not calibrated: physical calibration not validated
Silhouette measurement used for grading	No
Whole-fish bounding box	1605.5811 x 598.7439 px

Measurement	Value
Body bounding box	968.09 x 484.25 px
Whole-fish measured ROI	1606 x 600 px
Body measured ROI	969 x 485 px
Relative width proxy	0.302
Proxy measurement source	existing ROI measurements
Measurement status	Unreliable
Orientation	unknown
Measurement reliability	Unreliable: overlapping fish; clipped fish; orientation unconfirmed
Calibration status	Not Calibrated
Normal reference range	Not calibrated
Width abnormality	Unreliable
Class C supporting points	Not calibrated
Contributed to final grade	No: not calibrated
Actual combined-score contribution	0

C. Image Features and Measurements

Whole Fish ROI — Measured

Recorded field	Value
bbox	[1949.00146484375,489.7748107910156,3554.58251953125,1088.5186767578125]
status	measured
measurement_source	Model 1 fish crop rectangular ROI; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,0,1606,600]}]
values.mean_hue_deg	226.0088820293319
values.median_hue_deg	165.00000000000006
values.mean_saturation	0.36003861876215626
values.median_saturation	0.16666666666666666
values.mean_value	0.2627593460796523
values.median_value	0.30980392156862746
values.yellow_ratio_proxy	0.08106683271066832
values.brown_ratio_proxy	0.05995433789954338
values.dark_patch_ratio_proxy	0.4361446658364467
values.discoloration_coverage_proxy	0.5268088418430884
values.specular_ratio_proxy	0

Body ROI — Measured

Recorded field	Value
bbox	[2193.06,489,3161.15,973.25]

Recorded field	Value
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[244,0,1213,485]}]
values.mean_hue_deg	210.95344667961112
values.median_hue_deg	164.99999999999983
values.mean_saturation	0.2607884038758371
values.median_saturation	0.11235955056179768
values.mean_value	0.3099971357900453
values.median_value	0.3764705882352941
values.yellow_ratio_proxy	0.07426297703020437
values.brown_ratio_proxy	0.04178183481748641
values.dark_patch_ratio_proxy	0.3060908791080187
values.discoloration_coverage_proxy	0.38714797910482696
values.specular_ratio_proxy	0

Head ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded

Recorded field	Value
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Tail ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

D. Final Grading Explanation

Decision stage	Result
Original weighted grade	Ungraded
Geometric support	Not calibrated
Geometric contribution	not calibrated
Final combined AI grade	Ungraded
Width changed classification	No
Decision reason	Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Highest weighted grade support was 37.95%, below the 50.00% final verdict threshold.
Next step	Review the reason above and acquire a clear view with reliable Model 2 Body evidence; geometric support cannot bypass missing evidence.

Model, source and threshold configuration

Recorded field	Value
session_id	Not recorded
timestamp	2026-11-01T00:20:20.811Z
traceability.model2_id	yolo26m_partdet
traceability.model2_name	YOLO26m Part Detector (r03 devbest)
traceability.model2_checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt
traceability.model2_checkpoint_sha256	f74cd336b6ba
traceability.grading_config_version	4.0
traceability.part_weights.Body	0.5
traceability.part_weights.Head	0.3
traceability.part_weights.Tail	0.2
traceability.model2_detector_threshold	0.25
traceability.minimum_part_evidence_threshold	0.25
traceability.minimum_grade_margin	0
traceability.quality_threshold	0.25
traceability.final_verdict_threshold	0.5
traceability.minimum_original_weight_coverage	0.5
traceability.grading_mode	tail_optional_v1
traceability.quality_inference_mode	full_frame
traceability.roi_padding	0
traceability.model2_available	true
traceability.model1_checkpoint	model1_fish_parent_detector.pt
traceability.model1_checkpoint_sha256	87498c748cec
traceability.detection_threshold	0.5
grading_config.config_version	4.0
grading_config.part_weights.Head	0.3

Recorded field	Value
grading_config.part_weights.Body	0.5
grading_config.part_weights.Tail	0.2
grading_config.require_body	true
grading_config.minimum_part_confidence	0.25
grading_config.minimum_final_score	Not recorded
grading_config.deprecated_minimum_final_score_used	false
grading_config.final_verdict_threshold	0.5
grading_config.active_final_verdict_threshold	0.5
grading_config.grading_mode	tail_optional_v1
grading_config.minimum_regions_observed	2
grading_config.minimum_original_weight_coverage	0.5
grading_config.strict_minimum_regions_observed	3
grading_config.strict_minimum_original_weight_coverage	1
grading_config.minimum_grade_margin	0
grading_config.minimum_track_observations	2
grading_config.temporal_evidence_limit	120
grading_config.maximum_grade_stddev	Not recorded
grading_config.rejected_override_threshold	Not recorded
grading_config.color_adjustments_enabled	false
grading_config.defect_adjustments_enabled	false
grading_config.image_diagnostics_enabled	true
grading_config.frame_quality.use_frame_quality_filter	false
grading_config.frame_quality.minimum_detection_confidence	Not recorded
grading_config.frame_quality.minimum_crop_width	Not recorded
grading_config.frame_quality.minimum_crop_height	Not recorded
grading_config.frame_quality.minimum_crop_area	Not recorded
grading_config.frame_quality.reject_clipped_crops	false
grading_config.frame_quality.sharpness_filter_enabled	false
grading_config.frame_quality.minimum_sharpness	Not recorded
grading_config.best_frame.detection_weight	0.35
grading_config.best_frame.area_weight	0.2
grading_config.best_frame.sharpness_weight	0.25
grading_config.best_frame.region_weight	0.2
grading_config.best_frame.area_reference	Not recorded
grading_config.best_frame.sharpness_reference	Not recorded
grading_config.best_frame.only_usable	true
grading_config.best_frame.clipped_penalty	0
grading_config.best_frame.save_best_fish_crop	false

Recorded field	Value
grading_config.best_frame.save_annotated_best_fish_crop	false
grading_config.best_frame.best_crop_directory	results/best_fish_crops
width_threshold_configuration.scoring_enabled	false
width_threshold_configuration.calibration_validated	false
width_threshold_configuration.calibration_id	Not recorded
width_threshold_configuration.validation_sample_count	0
width_threshold_configuration.normal_reference_min	Not recorded
width_threshold_configuration.normal_reference_max	Not recorded
width_threshold_configuration.extreme_reference	Not recorded
width_threshold_configuration.scoring_method	post_weighted_v2
width_threshold_configuration.maximum_geometric_contribution	0.1
width_threshold_configuration.maximum_body_score_shift	0.1
width_threshold_configuration.minimum_class_c_evidence	0.25
width_threshold_configuration.geometry_max_axis_deviation_degrees	15
width_threshold_configuration.geometry_min_elongation	4
width_threshold_configuration.minimum_detection_confidence	0.25
width_threshold_configuration.minimum_part_inside_fraction	0.95

Fish #6 · Ungraded · Final support 5.2%

A. Grading Evidence

Evidence	Value
Model 1 detection	63.7%
Model 2 Body	Not Detected
Model 2 Head	Not Detected
Model 2 Tail	Class B · 26.0% part confidence
Effective weights	Head: 0 · Body: 0 · Tail: 0.2
Original weighted classification	Ungraded
Original weighted support	5.2%
Final combined support	5.2%
Ungraded reason	Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Reliable original anatomy-weight coverage was below 50.00%. Highest weighted grade support was 5.20%, below the 50.00% final verdict threshold.

Class	Original weighted support	Final combined support
Class A	0	0
Class B	0.051974356174468994	0.051974356174468994
Class C	0	0
Rejected	0	0

Model 2 label	Region	Class	Confidence (0–1)	Original bbox (px)
Grade_B_Tail	Tail	Class B	0.25987178087234497	[727.07,0,1009,657]

B. Body Width and Geometric Analysis

Measurement	Value
Measured fish length (px)	Unavailable
Maximum body width (px)	Unavailable
Relative body width ratio	Unavailable
Fish measurement status	unavailable
Fish measurement reliability	Unavailable: parent association uncertain; per fish segmentation unavailable; historical image unavailable
Measurement validation	Not validated for anatomical measurement or width grading
Body measurement source	Unavailable
Fish measurement method	segmentation centerline profile v1
Fish orientation (degrees)	Unavailable
Physical calibration	Not calibrated: physical calibration not validated
Silhouette measurement used for grading	No
Whole-fish bounding box	290.5063 x 656.7776 px

Measurement	Value
Body bounding box	Unavailable
Whole-fish measured ROI	292 x 657 px
Body measured ROI	Unavailable
Relative width proxy	Unavailable
Proxy measurement source	detector bounding boxes
Measurement status	Missing Detection
Orientation	unknown
Measurement reliability	Unavailable: body evidence unavailable; overlapping fish
Calibration status	Not Calibrated
Normal reference range	Not calibrated
Width abnormality	Unavailable
Class C supporting points	Not calibrated
Contributed to final grade	No: not calibrated
Actual combined-score contribution	0

C. Image Features and Measurements

Whole Fish ROI — Measured

Recorded field	Value
bbox	[717.7841186523438,0.0305938720703125,1008.2904052734375,656.8081665039062]
status	measured
measurement_source	Model 1 fish crop rectangular ROI; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,0,292,657]}]
values.mean_hue_deg	254.22399205326977
values.median_hue_deg	71.99999999999996
values.mean_saturation	0.43405387519767763
values.median_saturation	0.2413793103448276
values.mean_value	0.18492306453241625
values.median_value	0.14901960784313725
values.yellow_ratio_proxy	0.05083818102208044
values.brown_ratio_proxy	0.10787931861303976
values.dark_patch_ratio_proxy	0.6042461583369821
values.discoloration_coverage_proxy	0.6764662955317863
values.specular_ratio_proxy	0

Body ROI — Not Detected

Recorded field	Value
bbox	Not recorded

Recorded field	Value
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Head ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured

Recorded field	Value
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Tail ROI — Measured

Recorded field	Value
bbox	[727.07,0,1009,657]
status	measured
measurement_source	Model 2 part-detection bounding box; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[10,0,292,657]}]
values.mean_hue_deg	254.83581854530513
values.median_hue_deg	71.99999999999996
values.mean_saturation	0.427190699847417
values.median_saturation	0.2266666666666667
values.mean_value	0.18699717027478327
values.median_value	0.15294117647058825
values.yellow_ratio_proxy	0.04879259906948628
values.brown_ratio_proxy	0.10349536362360612
values.dark_patch_ratio_proxy	0.5991558448568067
values.discoloration_coverage_proxy	0.667481675788292
values.specular_ratio_proxy	0

D. Final Grading Explanation

Decision stage	Result
Original weighted grade	Ungraded
Geometric support	Not calibrated
Geometric contribution	not calibrated
Final combined AI grade	Ungraded
Width changed classification	No
Decision reason	Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Reliable original anatomy-weight coverage was below 50.00%. Highest weighted grade support was 5.20%, below the 50.00% final verdict threshold.
Next step	Review the reason above and acquire a clear view with reliable Model 2 Body evidence; geometric support cannot bypass missing evidence.

Model, source and threshold configuration

Recorded field	Value
session_id	Not recorded
timestamp	2026-11-01T00:20:20.811Z
traceability.model2_id	yolo26m_partdet
traceability.model2_name	YOLO26m Part Detector (r03 devbest)
traceability.model2_checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt
traceability.model2_checkpoint_sha256	f74cd336b6ba
traceability.grading_config_version	4.0
traceability.part_weights.Body	0.5
traceability.part_weights.Head	0.3
traceability.part_weights.Tail	0.2
traceability.model2_detector_threshold	0.25
traceability.minimum_part_evidence_threshold	0.25
traceability.minimum_grade_margin	0
traceability.quality_threshold	0.25
traceability.final_verdict_threshold	0.5
traceability.minimum_original_weight_coverage	0.5
traceability.grading_mode	tail_optional_v1
traceability.quality_inference_mode	full_frame
traceability.roi_padding	0
traceability.model2_available	true
traceability.model1_checkpoint	model1_fish_parent_detector.pt
traceability.model1_checkpoint_sha256	87498c748cec
traceability.detection_threshold	0.5
grading_config.config_version	4.0
grading_config.part_weights.Head	0.3

Recorded field	Value
grading_config.part_weights.Body	0.5
grading_config.part_weights.Tail	0.2
grading_config.require_body	true
grading_config.minimum_part_confidence	0.25
grading_config.minimum_final_score	Not recorded
grading_config.deprecated_minimum_final_score_used	false
grading_config.final_verdict_threshold	0.5
grading_config.active_final_verdict_threshold	0.5
grading_config.grading_mode	tail_optional_v1
grading_config.minimum_regions_observed	2
grading_config.minimum_original_weight_coverage	0.5
grading_config.strict_minimum_regions_observed	3
grading_config.strict_minimum_original_weight_coverage	1
grading_config.minimum_grade_margin	0
grading_config.minimum_track_observations	2
grading_config.temporal_evidence_limit	120
grading_config.maximum_grade_stddev	Not recorded
grading_config.rejected_override_threshold	Not recorded
grading_config.color_adjustments_enabled	false
grading_config.defect_adjustments_enabled	false
grading_config.image_diagnostics_enabled	true
grading_config.frame_quality.use_frame_quality_filter	false
grading_config.frame_quality.minimum_detection_confidence	Not recorded
grading_config.frame_quality.minimum_crop_width	Not recorded
grading_config.frame_quality.minimum_crop_height	Not recorded
grading_config.frame_quality.minimum_crop_area	Not recorded
grading_config.frame_quality.reject_clipped_crops	false
grading_config.frame_quality.sharpness_filter_enabled	false
grading_config.frame_quality.minimum_sharpness	Not recorded
grading_config.best_frame.detection_weight	0.35
grading_config.best_frame.area_weight	0.2
grading_config.best_frame.sharpness_weight	0.25
grading_config.best_frame.region_weight	0.2
grading_config.best_frame.area_reference	Not recorded
grading_config.best_frame.sharpness_reference	Not recorded
grading_config.best_frame.only_usable	true
grading_config.best_frame.clipped_penalty	0
grading_config.best_frame.save_best_fish_crop	false

Recorded field	Value
grading_config.best_frame.save_annotated_best_fish_crop	false
grading_config.best_frame.best_crop_directory	results/best_fish_crops
width_threshold_configuration.scoring_enabled	false
width_threshold_configuration.calibration_validated	false
width_threshold_configuration.calibration_id	Not recorded
width_threshold_configuration.validation_sample_count	0
width_threshold_configuration.normal_reference_min	Not recorded
width_threshold_configuration.normal_reference_max	Not recorded
width_threshold_configuration.extreme_reference	Not recorded
width_threshold_configuration.scoring_method	post_weighted_v2
width_threshold_configuration.maximum_geometric_contribution	0.1
width_threshold_configuration.maximum_body_score_shift	0.1
width_threshold_configuration.minimum_class_c_evidence	0.25
width_threshold_configuration.geometry_max_axis_deviation_degrees	15
width_threshold_configuration.geometry_min_elongation	4
width_threshold_configuration.minimum_detection_confidence	0.25
width_threshold_configuration.minimum_part_inside_fraction	0.95

Fish #7 · Ungraded · Final support -

A. Grading Evidence

Evidence	Value
Model 1 detection	59.5%
Model 2 Body	Not Detected
Model 2 Head	Not Detected
Model 2 Tail	Not Detected
Effective weights	Head: 0 · Body: 0 · Tail: 0
Original weighted classification	Ungraded
Original weighted support	-
Final combined support	-
Ungraded reason	Model 2 part-to-parent association is ambiguous, so no verdict was issued. Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Reliable original anatomy-weight coverage was below 50.00%. Highest weighted grade support was 0.00%, below the 50.00% final verdict threshold.

Class	Original weighted support	Final combined support
Class A	0	0
Class B	0	0
Class C	0	0
Rejected	0	0

Model 2 detections: Not Detected

B. Body Width and Geometric Analysis

Measurement	Value
Measured fish length (px)	Unavailable
Maximum body width (px)	Unavailable
Relative body width ratio	Unavailable
Fish measurement status	unreliable
Fish measurement reliability	Unreliable: parent association uncertain; parent touches frame boundary; per fish segmentation unavailable; historical image unavailable
Measurement validation	Not validated for anatomical measurement or width grading
Body measurement source	Unavailable
Fish measurement method	segmentation centerline profile v1
Fish orientation (degrees)	Unavailable
Physical calibration	Not calibrated: physical calibration not validated
Silhouette measurement used for grading	No
Whole-fish bounding box	645.7642 x 749.5273 px
Body bounding box	Unavailable

Measurement	Value
Whole-fish measured ROI	647 x 750 px
Body measured ROI	Unavailable
Relative width proxy	Unavailable
Proxy measurement source	detector bounding boxes
Measurement status	Missing Detection
Orientation	unknown
Measurement reliability	Unavailable: body evidence unavailable; association unreliable; overlapping fish; clipped fish
Calibration status	Not Calibrated
Normal reference range	Not calibrated
Width abnormality	Unavailable
Class C supporting points	Not calibrated
Contributed to final grade	No: not calibrated
Actual combined-score contribution	0

C. Image Features and Measurements

Whole Fish ROI — Measured

Recorded field	Value
bbox	[1183.9931640625,1410.47265625,1829.75732421875,2160]
status	measured
measurement_source	Model 1 fish crop rectangular ROI; no segmentation mask
missing_value_reason	Not recorded
frame_rois	[{"frame_id":"still-full-frame","frame_id_missing_reason":null,"crop_local_bbox":[0,0,647,750]}]
values.mean_hue_deg	225.21642079944854
values.median_hue_deg	173.33333333333337
values.mean_saturation	0.4731234547233001
values.median_saturation	0.25373134328358216
values.mean_value	0.22461225767999113
values.median_value	0.25098039215686274
values.yellow_ratio_proxy	0.08673261205564142
values.brown_ratio_proxy	0.05069345698093766
values.dark_patch_ratio_proxy	0.5058423493044822
values.discoloration_coverage_proxy	0.5945656877897991
values.specular_ratio_proxy	0

Body ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable

Recorded field	Value
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Head ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured

Recorded field	Value
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

Tail ROI — Not Detected

Recorded field	Value
bbox	Not recorded
status	unavailable
measurement_source	Not recorded
missing_value_reason	No Model 2 part detection in this frame
missing_values.mean_hue_deg	undefined_or_not_measured
missing_values.median_hue_deg	undefined_or_not_measured
missing_values.mean_saturation	undefined_or_not_measured
missing_values.median_saturation	undefined_or_not_measured
missing_values.mean_value	undefined_or_not_measured
missing_values.median_value	undefined_or_not_measured
missing_values.yellow_ratio_proxy	undefined_or_not_measured
missing_values.brown_ratio_proxy	undefined_or_not_measured
missing_values.dark_patch_ratio_proxy	undefined_or_not_measured
missing_values.discoloration_coverage_proxy	undefined_or_not_measured
missing_values.specular_ratio_proxy	undefined_or_not_measured
frame_rois	[]
values.mean_hue_deg	Not recorded
values.median_hue_deg	Not recorded

Recorded field	Value
values.mean_saturation	Not recorded
values.median_saturation	Not recorded
values.mean_value	Not recorded
values.median_value	Not recorded
values.yellow_ratio_proxy	Not recorded
values.brown_ratio_proxy	Not recorded
values.dark_patch_ratio_proxy	Not recorded
values.discoloration_coverage_proxy	Not recorded
values.specular_ratio_proxy	Not recorded

D. Final Grading Explanation

Decision stage	Result
Original weighted grade	Ungraded
Geometric support	Not calibrated
Geometric contribution	not calibrated
Final combined AI grade	Ungraded
Width changed classification	No
Decision reason	Model 2 part-to-parent association is ambiguous, so no verdict was issued. Required Body and Head evidence was not reliably observed; this is not a physical missing-part claim. Fewer than 2 reliable regions were observed. Reliable original anatomy-weight coverage was below 50.00%. Highest weighted grade support was 0.00%, below the 50.00% final verdict threshold.
Next step	Review the reason above and acquire a clear view with reliable Model 2 Body evidence; geometric support cannot bypass missing evidence.

Model, source and threshold configuration

Recorded field	Value
session_id	Not recorded
timestamp	2026-11-01T00:20:20.811Z
traceability.model2_id	yolo26m_partdet
traceability.model2_name	YOLO26m Part Detector (r03 devbest)
traceability.model2_checkpoint	model2_tamban_yolo26m_partdet_r03_devbest.pt
traceability.model2_checkpoint_sha256	f74cd336b6ba
traceability.grading_config_version	4.0
traceability.part_weights.Body	0.5
traceability.part_weights.Head	0.3
traceability.part_weights.Tail	0.2
traceability.model2_detector_threshold	0.25
traceability.minimum_part_evidence_threshold	0.25
traceability.minimum_grade_margin	0
traceability.quality_threshold	0.25

Recorded field	Value
traceability.final_verdict_threshold	0.5
traceability.minimum_original_weight_coverage	0.5
traceability.grading_mode	tail_optional_v1
traceability.quality_inference_mode	full_frame
traceability.roi_padding	0
traceability.model2_available	true
traceability.model1_checkpoint	model1_fish_parent_detector.pt
traceability.model1_checkpoint_sha256	87498c748cec
traceability.detection_threshold	0.5
grading_config.config_version	4.0
grading_config.part_weights.Head	0.3
grading_config.part_weights.Body	0.5
grading_config.part_weights.Tail	0.2
grading_config.require_body	true
grading_config.minimum_part_confidence	0.25
grading_config.minimum_final_score	Not recorded
grading_config.deprecated_minimum_final_score_used	false
grading_config.final_verdict_threshold	0.5
grading_config.active_final_verdict_threshold	0.5
grading_config.grading_mode	tail_optional_v1
grading_config.minimum_regions_observed	2
grading_config.minimum_original_weight_coverage	0.5
grading_config.strict_minimum_regions_observed	3
grading_config.strict_minimum_original_weight_coverage	1
grading_config.minimum_grade_margin	0
grading_config.minimum_track_observations	2
grading_config.temporal_evidence_limit	120
grading_config.maximum_grade_stddev	Not recorded
grading_config.rejected_override_threshold	Not recorded
grading_config.color_adjustments_enabled	false
grading_config.defect_adjustments_enabled	false
grading_config.image_diagnostics_enabled	true
grading_config.frame_quality.use_frame_quality_filter	false
grading_config.frame_quality.minimum_detection_confidence	Not recorded
grading_config.frame_quality.minimum_crop_width	Not recorded
grading_config.frame_quality.minimum_crop_height	Not recorded
grading_config.frame_quality.minimum_crop_area	Not recorded
grading_config.frame_quality.reject_clipped_crops	false

Recorded field	Value
grading_config.frame_quality.sharpness_filter_enabled	false
grading_config.frame_quality.minimum_sharpness	Not recorded
grading_config.best_frame.detection_weight	0.35
grading_config.best_frame.area_weight	0.2
grading_config.best_frame.sharpness_weight	0.25
grading_config.best_frame.region_weight	0.2
grading_config.best_frame.area_reference	Not recorded
grading_config.best_frame.sharpness_reference	Not recorded
grading_config.best_frame.only_usable	true
grading_config.best_frame.clipped_penalty	0
grading_config.best_frame.save_best_fish_crop	false
grading_config.best_frame.save_annotated_best_fish_crop	false
grading_config.best_frame.best_crop_directory	results/best_fish_crops
width_threshold_configuration.scoring_enabled	false
width_threshold_configuration.calibration_validated	false
width_threshold_configuration.calibration_id	Not recorded
width_threshold_configuration.validation_sample_count	0
width_threshold_configuration.normal_reference_min	Not recorded
width_threshold_configuration.normal_reference_max	Not recorded
width_threshold_configuration.extreme_reference	Not recorded
width_threshold_configuration.scoring_method	post_weighted_v2
width_threshold_configuration.maximum_geometric_contribution	0.1
width_threshold_configuration.maximum_body_score_shift	0.1
width_threshold_configuration.minimum_class_c_evidence	0.25
width_threshold_configuration.geometry_max_axis_deviation_degrees	15
width_threshold_configuration.geometry_min_elongation	4
width_threshold_configuration.minimum_detection_confidence	0.25
width_threshold_configuration.minimum_part_inside_fraction	0.95

Rectangular ROI colour measurements are diagnostics and may include background. Physical dimensions require an applied, validated calibration matching the source and frame. Anatomical measurement validation is separate from pixel geometry reliability. Original AI verdicts remain separate from operator reviews.